→ Decision ... Numerical example

Day	outlook	temp	Humidity	wind	play tennis
D1	Sunny	Hot	High	weak	no
D2	Sunny	Hot	High	strong	no
D3	overcast	Hot	high	weak	yes
D4	Rain	mild	High	weak	yes
D5	Rain	cool	normal	weak	yes
D6	Rain	cool	normal	strong	no
D7	overcast	cool	normal	strong	yes
D8	sunny	mild	high	weak	no
D9	sunny	cool	normal	weak	yes
D10	Rain	mild	normal	weak	yes
D11	sunny	mild	normal	strong	yes
D12	overcast	mild	high	strong	yes
D13	overcast	Hot	normal	weak	yes
D14	Rain	mild	High	strong	no

this is Data set consider Decision tree.
14 numerical examples, 4 attributes.

→ we need to understand which one is maximal information i.e rootnode

Attribute: out look

values (outlook) = Sunny, overcast, Rain

whole data set $S = [9+, 5-]$

$$\text{entropy}(S) = -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14} = 0.94$$

entire data set (Proposition)

S_{sunny} [2+, 3−]

$$\text{entropy}(S_{sunny}) = -\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5}$$

$$= 0.971$$

entropy $S_{overcast}$ [4+, 0−]

$$\text{entropy}(S_{overcast}) = -\frac{4}{4}\log_2\frac{4}{4} - \frac{0}{4}\log_2\frac{0}{4}$$

$$= 0$$

S_{Rain} [3+, 2−] =

$$\text{entropy}(S_{Rain}) = -\frac{3}{5}\log_2\frac{3}{5} - \frac{2}{5}\log_2\frac{2}{5}$$

$$= 0.971.$$

$$\text{Gain}(S, \text{outlook}) = \text{Entropy}(S) - \sum_{v \in \{\text{sunny, overcast, Rain}\}} \frac{|S_v|}{|S|}$$

S_v is a cardinality of S_v

$|S|$ is cardinality

Gain (S, outlook)

$$= \text{Entropy}(S) - \frac{5}{14}\text{Entropy}(S_{sunny}) - \frac{4}{14}\text{Entropy}$$

$$\frac{5}{14}\text{Entropy}(S_{Rain})$$

Gain (S, outlook)

$$= 0.94 - \frac{5}{14} * 0.971 - \frac{4}{14} * 0 - \frac{5}{14} * 0.971$$

Gain(S, outlook) = 0.2464

Attribute : Temp

values (Temp) = Hot, mild, cool

$S = [9+, 5-]$

$$\text{Entropy}(S) = -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14} = 0.94.$$

S_{Hot} $[2+, 2-]$

$$\text{Entropy}(S_{Hot}) = -\frac{2}{4}\log_2\frac{2}{4} - \frac{2}{4}\log_2\frac{2}{4}$$

$$= 1.0.$$

S_{mild} $[4+, 2-]$

$$\text{Entropy}(S_{mild}) = -\frac{4}{6}\log_2\frac{4}{6} - \frac{2}{6}\log_2\frac{2}{6}$$

$$= 0.9183.$$

S_{cool} $[3+, 1-]$

entropy(Sv)

$$\text{entropy}(S_{cool}) = -\frac{3}{4}\log_2\frac{3}{4} - \frac{1}{4}\log_2\frac{1}{4}$$

$$= 0.8113$$

$$\text{Gain}(S, \text{temp}) = \text{Entropy}(S) - \sum_{v \in (\text{Hot}, \text{mild}, \text{cool})} \frac{|S_v|}{|S|}\text{entropy}(S_v)$$

(Sover cost) =

$$= \text{Entropy}(S) - \frac{4}{14}\text{Entropy}(S_{Hot}) - \frac{6}{14}\text{Entropy}(S_{mild}) - \frac{4}{14}\text{Entropy}(S_{cool})$$

$$\text{Gain}(S, \text{Temp}) = 0.94 - \frac{4}{14} \times 1.0 - \frac{6}{14} \times 0.9183 - \frac{4}{14} \times 0.8113 = 0.0289$$

Attribute : Humidity

values (Humidity) = High, normal

$S = [9+, 5-]$

$$\text{entropy}(S) = -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14}$$

$$= 0.94$$

S_{high} $[3+, 4-]$

$$\text{entropy}(S_{high}) = -\frac{3}{7}\log_2\frac{3}{7} - \frac{4}{7}\log_2\frac{4}{7}$$

$$= 0.9852$$

S_{normal} $[6+, 1-]$

$$\text{entropy}(S_{normal}) = -\frac{6}{7}\log_2\frac{6}{7} - \frac{1}{7}\log_2\frac{1}{7}$$

$$= 0.5916$$

$$\text{Gain}(S, \text{Humidity}) = \text{Entropy}(S) - \sum_{v \in \{high, normal\}} \frac{|S_v|}{|S|} \text{entr}$$

$$\text{Gain}(S, \text{Humidity})$$

$$= \text{Entropy}(S) - \frac{7}{14}\text{entropy}(S_{high}) - \frac{7}{14}\text{Entropy}($$

$$= 0.94 - \frac{7}{14} * 0.9852 - \frac{7}{14} * 0.5916$$

$$\text{Gain}(S, \text{Humidity}) = 0.1516.$$

Attribute : wind

values(wind) = strong, weak

$S = [9+, 5-]$

$$entropy(S) = -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14} = 0.94$$

S_{strong} [3+, 3-]

$entropy(S_{strong}) = 1.0$

S_{weak} (6+, 2-]

$$entropy(S_{weak}) = -\frac{6}{8}\log_2\frac{6}{8} - \frac{2}{8}\log_2\frac{2}{8}$$

$= 0.8113.$

$$Gain(S, wind) = Entropy(S) - \sum_{v \in \{strong, weak\}} \frac{|S_v|}{|S|} entropy(S_v)$$

py (Sv)

$$Gain(S, wind) = Entropy(S) - \frac{6}{14} Entropy(S_{strong}) - \frac{8}{14} Entropy(S_{weak})$$

normal)

$$Gain(S, wind) = 0.94 - \frac{6}{14} * 1.0 - \frac{8}{14} * 0.8113$$

$= 0.0478.$

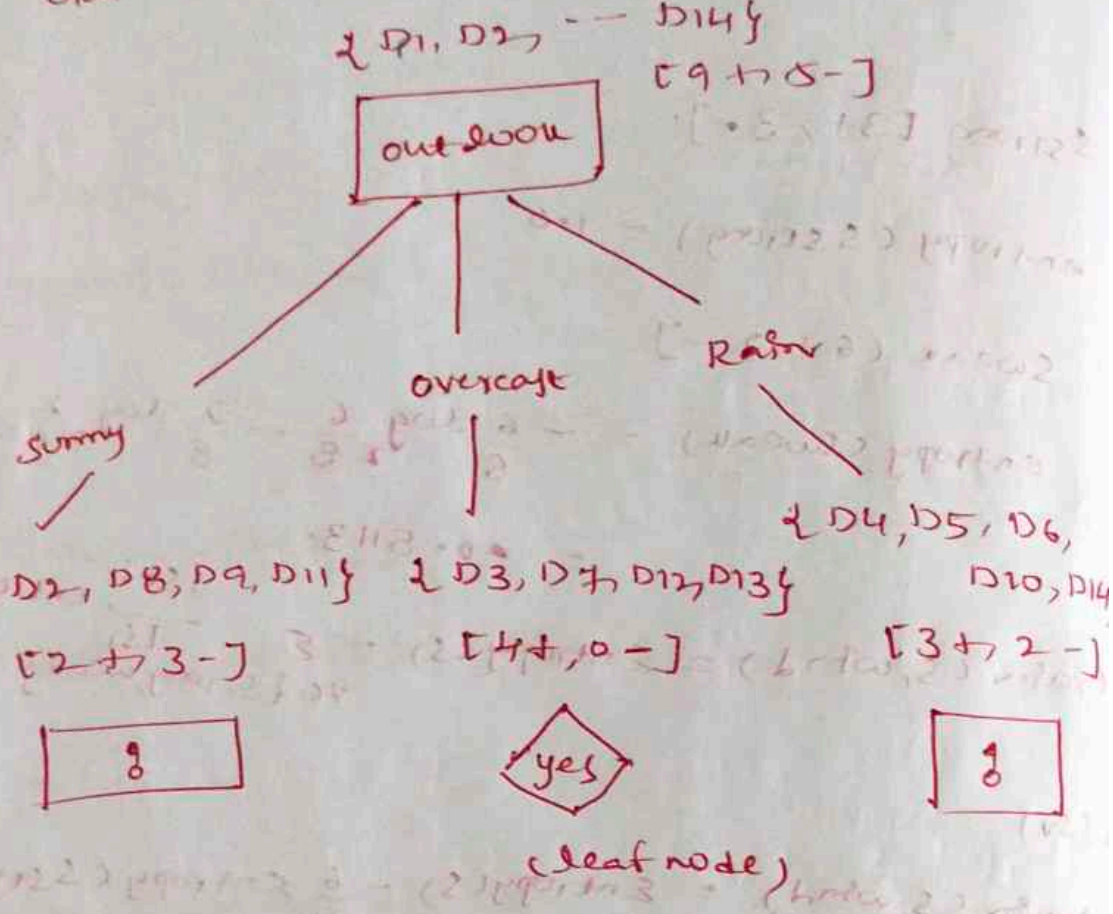
Gain of every attribute

Gain (S, outlook) = 0.2464 → maximum Gain → root node

Gain (S, Temp) = 0.0289

Gain (S, Humidity) = 0.1516

Gain (S, wind) = 0.0478.



Day	Temp	Humidity	wind	play Tennis
D1	Hot	High	weak	no
D2	Hot	High	Strong	no
D8	mild	High	weak	no
D9	cool	normal	weak	yes
D11	mild	normal	Strong	yes.

Attribute : Temp

values (Temp) = Hot, mild, cool.

$S_{sunny} = [2+, 3-]$

$Entropy(S_{sunny}) = -\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5}$

$= 0.97$

$S_{Hot}[0+, 2-]$

$Entropy(S_{Hot}) = 0.0$

$S_{mild}[1+, 1-]$ $Entropy(S_{mild}) = 1.0$

$S_{cool}[1+, 0-]$ $Entropy(S_{cool}) = 0.0$

$Gain(S_{sunny}, Temp) = Entropy(S) - \sum_{v \in \{Hot, mild, cool\}} \frac{|S_v|}{|S|} entropy(S_v)$

$Gain(S_{sunny}, Temp)$

$= Entropy(S) - \frac{2}{5} Entropy(S_{Hot}) - \frac{2}{5} entropy(S_{mild}) - \frac{1}{5} entropy(S_{cool})$

$= 0.97 - \frac{2}{5} * 0.0 - \frac{2}{5} * 1 - \frac{1}{5} * 0.0 = 0.570$

Attribute : Humidity

values (Humidity) = High, normal

$S_{sunny}[2+, 3-]$ $Entropy(S) = -\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5} = 0.97$

$S_{high}[0+, 3-]$ $entropy(S_{high}) = 0.0$

$S_{normal}[2+, 0-]$ $entropy(S_{normal}) = 0.0$

$Gain(S_{sunny}, Humidity) = entropy(S) - \sum_{v \in \{High, normal\}} \frac{|S_v|}{|S|}$

$Gain(S_{sunny}, Humidity) = entropy(S) - \frac{3}{5} entropy($

$= 0.97 - \frac{3}{5} * 0 - \frac{2}{5} * 0$

$= 0.97$

Attribute : wind

values (wind) = strong, weak

$S_{sunny} = [2+, 3-]$

$entropy(S) = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.97$

S_{strong} [1+, 1-]

$entropy(S_{strong}) = 1.0$

S_{weak} [1+, 2-]

$entropy(S_{weak}) = -\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 2/3$

$= 0.9183.$

$Gain(S_{sunny}, wind) = Entropy(S) - \sum_{v \in \{strong, weak\}} \frac{|S_v|}{|S|} entropy$

$Gain(S_{sunny}, wind) = Entropy(S) - \frac{2}{5} entropy(S_{strong})$

$= 0.97 - \frac{2}{5} * 1.0 - \frac{3}{5} * 0.918$

$= 0.0192$

entropy (Sv)

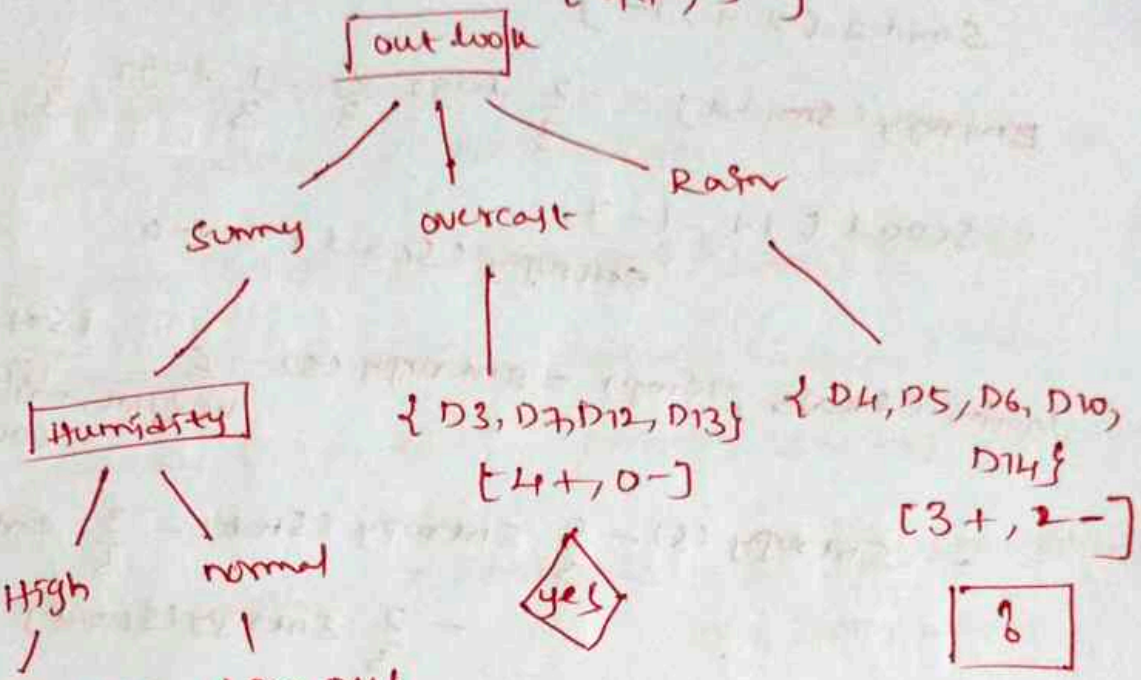
high) − $\frac{2}{5}$ Entropy (Snormal)

Gain (Ssunny, Temp) = 0.570

Gain (Ssunny, Humidity) = 0.97 → highest

Gain (Ssunny, wind) = 0.0192

{D1, D2 ... D14}
[9+, 5−]



(Sv)

Day	Temp	Humidity	wind	play Tennis
D4	mild	High	weak	yes
D5	cool	normal	weak	yes
D6	cool	normal	strong	no
D10	mild	normal	weak	yes
D14	mild	High	strong	no

− $\frac{3}{5}$ Entropy (Sweak)

3

Attribute: Temp

values (Temp) = Hot, mild, cool

S_{Rain} [3+, 2−]

$$\text{entropy}(S_{sunny}) = -\frac{3}{5}\log_2\frac{3}{5} - \frac{2}{5}\log_2\frac{2}{5} = 0.97$$

S_{Hot} [0+, 0−] entropy(S_{hot}) = 0.0

S_{mild} [2+, 1−]

$$\text{Entropy}(S_{mild}) = -\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3} = 0.918$$

S_{cool} [1+, 1−]

entropy(S_{cool}) = 1.0.

$$\text{Gain}(S_{Rain}, \text{Temp}) = \text{Entropy}(S) - \sum_{v \in \{\text{Hot, mild, cool}\}} \frac{|S_v|}{|S|} \text{Entropy}$$

$$= \text{Entropy}(S) - \frac{0}{5}\text{Entropy}(S_{hot}) - \frac{3}{5}\text{entropy}($$

$$- \frac{2}{5}\text{Entropy}(S_{cool})$$

$$= 0.97 - \frac{0}{5} * 0.0 - \frac{3}{5} * 0.918 - \frac{2}{5} * 1.0$$

$$= 0.0192$$

Attribute: Humidity

values (Humidity) = high, normal

S_{Rain} [3+, 2−] entropy(S_{sunny}) = 0.97

S_{high} [1+, 1−] entropy(S_{high}) = 1.0

S_{normal} [2+, 1−] entropy(S_{normal})

$$= -\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3}$$

$$= 0.9183.$$

$$Gain(S_{Rain}, Humidity) = entropy(S) - \sum_{v \in \{high, normal\}} \frac{|S_v|}{|S|} * entropy(S_v)$$

$$= entropy(S) - \frac{2}{5} entropy(S_{high}) - \frac{3}{5} entropy(S_{normal})$$

$$= 0.97 - \frac{2}{5} * 1.0 - \frac{3}{5} * 0.918$$

$$= 0.0192$$

Attribute: wind

values (wind) = strong, weak

S_{Rain} [3+, 2−] Entropy (S_{sunny}) = 0.97

S_{strong} [0+, 2−] entropy (S_{strong}) = 0.0

S_{weak} [3+, 0−] entropy (S_{weak}) = 0.0

$$Gain(S_{Rain}, wind) = Entropy(S) - \sum_{v \in \{strong, weak\}} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$= Entropy(S) - \frac{2}{5} Entropy(S_{strong}) - \frac{3}{5} Entropy(S_{weak})$$

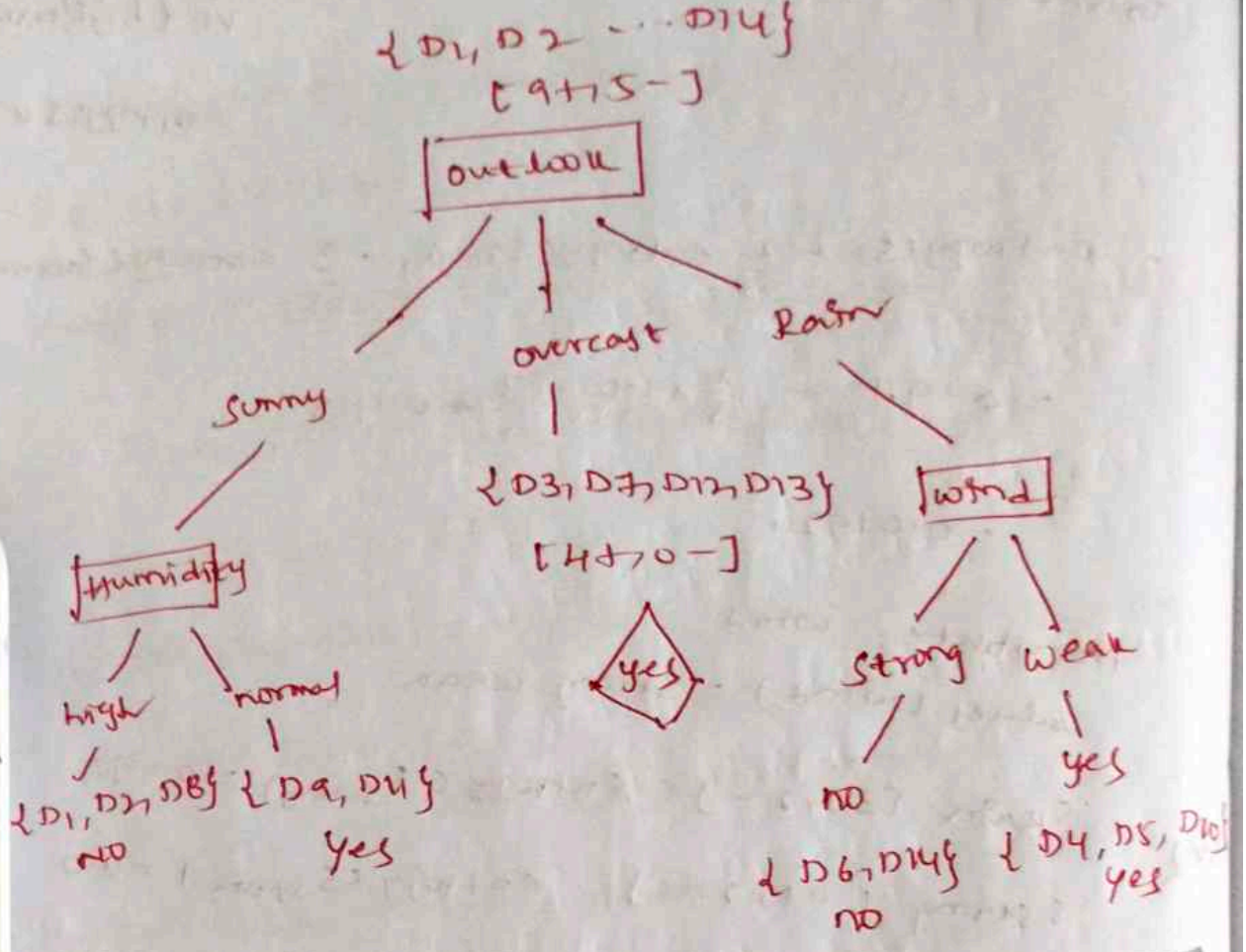
$$= 0.97 - \frac{2}{5} * 0.0 - \frac{3}{5} * 0.0$$

$$= 0.97$$

$Gain(S_{Rain}, Temp) = 0.0192$

$Gain(S_{Rain}, Humidity) = 0.0192$

$Gain(S_{Rain}, wind) = 0.97$ → more


{D1, D2 ... D14}
[9+, 5-]
outlook
sunny
overcast
Rain
{D3, D7, D12, D13}
[4+, 0-]
yes
humidity
high
normal
{D1, D2, D8}
NO
{D9, D11}
yes
wind
strong
weak
NO
{D6, D14}
NO
yes
{D4, D5, D10}
yes